

THE OUTER PLANETS

UNLIKE THE ROCKY COMPOSITION OF THE INNER PLANETS, THE FOUR GIANT OUTER PLANETS ARE LARGELY MADE OF GAS. THESE MASSIVE BODIES HAVE DOZENS OF MOONS, EACH FORMING A MINIATURE PLANETARY SYSTEM.

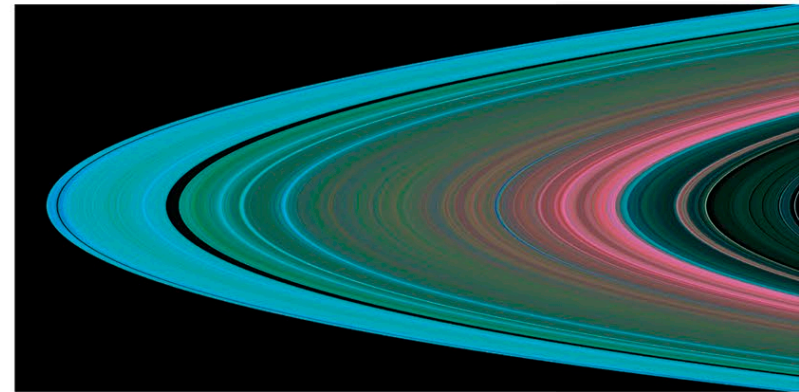
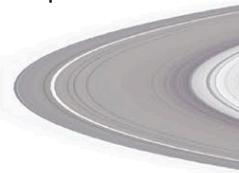
All four outer planets are much larger than Earth. Despite their size, they all have days shorter than Earth's. This rapid rotation leads to their atmosphere splitting up into bands. Visits by spacecraft have shown that they all possess a magnetic field, and aurorae occur in their atmospheres. The four bodies all have ring systems, but Saturn has the most extensive by far.

Jupiter's bands alternate between light-colored regions of rising air and darker regions of falling air



Saturn

Measuring over 75,898 miles (120,536 km) across, Saturn is the second-largest planet. It has a banded atmosphere similar to Jupiter's, but its cloud structures are more muted. The planet is surrounded by an extensive ring system, probably the remains of a destroyed moon. Its largest moon, Titan, has a thick atmosphere with a surface pressure higher than that at Earth's surface.

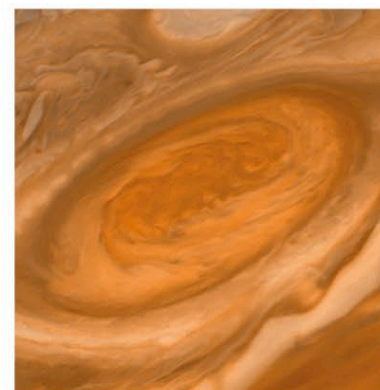


△ Saturn's rings

Saturn's rings are composed of chunks of almost pure water ice, with a tiny amount of dust. One of the largest rings, called the E ring, is still fed by ice grains erupting from the moon Enceladus.

Jupiter

Measuring 88,846 miles (142,984 km) across, its equator could fit ten Earths side-by-side. The largest planet's great mass affects the orbits of many bodies, and numerous comets' orbits have been altered by its presence. This vast planet has an extremely strong magnetic field. High energy particles are trapped by this field, making it a dangerous place for human exploration.



◁ Giant Red Spot

This spectacular, churning storm has existed in Jupiter's atmosphere for at least 300 years. Typically measuring 18,600 miles (30,000 km) across, Earth would fit inside it two or three times. It varies in shape and darkness, and its red color is probably due to chemicals being drawn up from deeper in the atmosphere.

▷ Europa

This large moon of Jupiter is a world of ice. Europa's cracked surface covers a global ocean of water heated by the effects of tides. This combination of water and heat means it's one of the few places where life could have arisen.

